Official Distrust, Investor Composition, and Sovereign Rollover Crises

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Abstract

The same fiscal report can support different rollover thresholds when different creditors hold the maturing debt. I study a global game in which creditor types differ in payoff hurdles and in effective pessimism about official information. A type-k creditor lowers the public signal by δ_k, which raises its run cutoff by $(\alpha/\beta)\delta_k$. Investor composition moves the unique crisis threshold through type-specific run propensities, amplified by the slope of the aggregate run map. Across 1,836 unique-equilibrium states, the coordination multiplier has median 1.56 and a 10th to 90th percentile range from 1.17 to 2.66. Official precision stabilizes after favorable news and destabilizes after adverse news. Credibility lowers the threshold at every announcement. Max-min ambiguity and Bayesian pessimistic bias generate identical equilibrium choices and prices. The empirical design therefore measures effective pessimism with revision histories and investor surveys, estimates payoff hurdles from creditor behavior, and tests their interaction in spreads and auctions. The reference calculation produces spread increments of 18 basis points in calm news and 98 basis points in stress.

JEL codes: D81, D82, F34, G01, H63.
Keywords: rollover risk, global games, ambiguity, multiple priors, investor base, transparency.

1 Introduction

In October 2009 the incoming Greek government announced that the budget deficit was far larger than previously reported. Subsequent releases raised the estimate again. The European Commission's January 2010 report documented serious weaknesses in Greek fiscal statistics (European Commission, 2010). The episode changed the information problem facing a creditor. Creditors now priced an unknown bias alongside ordinary sampling error in each published fiscal number.

This paper studies that doubt and the investor base that carries it. Two sovereigns can publish the same fiscal report and face different rollover thresholds because their maturing bonds sit with creditors who have different incentives to run. Creditors face an unknown bias in official information and evaluate rollover under the worst prior in an interval, following Gilboa and

Schmeidler (1989). Each type has its own rollover payoff, exit payoff, and effective pessimism. The investor base determines how much maturing debt each type controls.

The model has an exact as-if representation. Type k behaves like a Bayesian creditor who lowers the official signal by δ_k. Its private-signal cutoff rises by $(\alpha/\beta)\delta_k$. Distrust matters most where creditors place greater posterior weight on official information. The same representation proves that max-min ambiguity and Bayesian pessimistic bias produce identical market outcomes. The paper therefore centers its empirical analysis on effective pessimism and reserves the preference interpretation for evidence outside prices.

The global-game precision bound continues to select a unique equilibrium (Hellwig, 2002; Morris and Shin, 2003). Distrust moves each type's cutoff without changing the slope bound. Investor composition then moves a unique threshold. The derivative is the difference between the types' run probabilities, multiplied by the coordination term $1/(1-\Phi')$. The same mechanism makes a given composition shift more costly when the run margin is densely populated.

Official precision and official credibility move different margins. Precision places more weight on the announcement. It lowers the crisis threshold when the announcement is sufficiently favorable and raises it after adverse news. Credibility reduces δ_k and lowers the threshold at every announcement. The model gives the exact news boundary at which the precision effect changes sign.

The reference calculation produces an effective-pessimism spread increment of 18 basis points in calm news and 98 basis points in stress. Moving maturing debt toward the high-run-propensity type produces a 10 basis point gap before adverse news and a 44 basis point gap after it. Raising recovery from 0.60 to 0.70 lowers the stress spread from 575 to 128 basis points. The multiplier distribution shows how these magnitudes change across the unique-equilibrium state space.

Global games supply the equilibrium machinery (Hellwig, 2002; Morris and Shin, 2003; Morris and Shin, 2004). Multiple-priors preferences follow Gilboa and Schmeidler (1989). Ui (2025) shows that ambiguous-quality information can trigger a debt rollover crisis through strategic ambiguity. The present paper takes a different margin. It introduces heterogeneous effective pessimism and type-specific payoff hurdles, then lets investor composition move the aggregate run map. This structure yields exact composition derivatives, separates precision from credibility, and proves the observational equivalence that determines the empirical sequence. Sovereign rollover models provide the debt setting (Calvo, 1988; Cole and Kehoe, 2000; Bocola and Dovis, 2019). Holder accounts measure type weights while creditor behavior identifies their run propensities (Arslanalp and Tsuda, 2014).

The empirical sequence begins with information and creditor behavior. Real-time revision histories measure the objective error distribution. Investor surveys and release responses map that history into subjective δ_k. A hierarchical forecast model separates private-signal noise from persistent model heterogeneity and rounding. Current holder accounts measure composition, while flows and auction participation identify type-specific payoff hurdles. Spreads and auction outcomes then test the predicted interaction. Institutional events provide descriptive validation, and the triple interaction supplies the core market test.

2 The Rollover Game

The analytical structure is exact inside the Gaussian one-shot game. Propositions 2 through 6 follow from normal signals, max-min preferences over a bias interval, Assumption 1, and the uniqueness inequality. Section 6 attaches magnitudes to those objects with documented payoffs, investor-base shares, information precisions, and revision-based distrust measures. Section 5 chains the static game into the policy experiments.

2.1 Environment

A sovereign has a unit continuum of short-term creditors, each holding one maturing bond. Each creditor chooses to roll over or to run. The sovereign survives the rollover iff the mass of runners is at most its liquid capacity θ: fundamentals are the fraction of maturing debt the sovereign could redeem using buffers and primary surplus. If $\theta < 0$ the sovereign enters crisis regardless, and if $\theta > 1$ it survives even a full run. Between, survival is a coordination outcome.

Payoffs may differ across creditor types. Rolling over gives R_k if the sovereign survives and $\mathcal{R}_k$ in crisis. Exiting gives Q_k in survival and q_k in crisis. Assume $R_k \geq \mathcal{R}_k$, $Q_k \geq q_k$, and

$$R_k - \mathcal{R}_k > Q_k - q_k, \tag{1}$$

so rollover is more exposed to the crisis state than exit. Type k rolls over when its assessed crisis probability satisfies

$$(1-P)R_k + P\mathcal{R}_k \geq (1-P)Q_k + Pq_k \quad \Longleftrightarrow \quad P \leq \bar{p}_k \equiv \frac{R_k - Q_k}{(R_k - \mathcal{R}_k) - (Q_k - q_k)}. \tag{2}$$

The safe-exit benchmark sets $Q_k = q_k = 1 - \kappa_k$. The quantitative section further imposes common payoffs, giving $\bar{p} = 27.3$ percent.

Information has a public and a private layer. All creditors see the official signal $y = \theta + \sigma_y \eta$, precision $\alpha = \sigma_y^{-2}$: the budget documents and the program reviews. Each creditor i also has private information $x_i = \theta + \sigma_x \varepsilon_i$, precision $\beta = \sigma_x^{-2}$, with ε_i independent standard normal: his own analysis and client flows.

The equilibrium formulas use the diffuse-prior limit standard in the Gaussian global game. Conditional on (x_i, y), posterior precision is $\alpha + \beta$. A sequence of proper normal priors with variance tending to infinity gives the same formulas. Section 4.1 evaluates ex ante welfare under a proper objective distribution.

2.2 Distrust as multiple priors

The departure is that creditors discount the official signal. Type-k creditors entertain the set of models in which the official signal carries an unknown bias b,

$$y = \theta + b + \sigma_y \eta, \qquad b \in [-\delta_k,\, \delta_k],$$

and evaluate actions by the worst model in the set (Gilboa and Schmeidler, 1989). The max-min creditor asks what rollover is worth when official information overstates capacity by the largest amount it regards as plausible. The radius δ_k measures this subjective effective pessimism. Historical revisions describe realized reporting errors. Investor surveys and behavior around releases determine which part of that error distribution enters the creditor's model set. Section 7 estimates the mapping from revision histories and direct beliefs into δ_k.

The relevant restriction compares the state exposure of rollover with the state exposure of exit.

Assumption 1 (Common worst-case ordering)**.** *Conditional payoffs* $(R_k, \mathcal{R}_k, Q_k, q_k)$ *do not depend directly on the signal-bias model. Survival weakly raises the payoff to either action. Inequality* (1) *holds and* $0 < \bar{p}_k < 1$.

Both actions can now have state-dependent payoffs. A larger perceived crisis probability lowers the value of rollover more than the value of exit, so the same pessimistic model governs the comparison.

Lemma 1 (Worst-case selection and as-if pessimism)**.** *Under Assumption 1, the crisis event in a threshold equilibrium is* $\{\theta \le \theta^*\}$*. The worst-case model for both actions is* $b = \delta_k$*. Type* k *behaves as a Bayesian creditor who observed* $y - \delta_k$ *and uses the type-specific hurdle* $\bar{p}_k$*.*

Proposition 1 (Observational equivalence)**.** *Conditional on* $(\delta_k, \bar{p}_k)$*, creditor choices, the run mass, the crisis threshold, and model-implied spreads are identical under max-min ambiguity and under a Bayesian model in which type* k *believes the official signal has pessimistic bias* δ_k*. Market outcomes identify the common effective-pessimism representation. Evidence outside those outcomes identifies its preference source.*

Revision histories establish the support of past reporting errors. Direct belief evidence identifies the admissible bias set before market outcomes evaluate the coordination mechanism. The model's distinct observable is the heterogeneous response to a common announcement after separate measurements identify payoff hurdles and prior bias beliefs.

2.3 The effective-pessimism shift and uniqueness

Bayesian updating from the effective signal gives type k's posterior $\theta \mid x_i \sim N(m_k(x_i), (\alpha+\beta)^{-1})$. Threshold strategies are cutoffs x_k^*. The indifference condition $\mathbb{P}(\theta \le \theta^* \mid x_k^*) = \bar{p}_k$ gives

$$x_k^* = x_{B,k}^*(\theta^*) + \frac{\alpha}{\beta}\delta_k, \tag{3}$$

where $x_{B,k}^*$ contains the type-specific quantile $\Phi^{-1}(\bar{p}_k)$.

Proposition 2 (The effective-pessimism shift and its loading)**.** *Type* k*'s run cutoff exceeds its Bayesian cutoff by* $(\alpha/\beta)\delta_k$*. This loading is independent of* $\bar{p}_k$*, though the level of the cutoff is not.*

The loading α/β carries the economics. Distrust of the official source matters only insofar as the posterior leans on that source: where private information is abundant (β large), creditors cross-check the government and δ is nearly free. Where the official account is all there is, every unit of doubt translates into cutoff. This is a cross-country prediction, stated as P2 in Section 7, and it inverts a comfortable intuition: the sovereigns most exposed to a credibility loss are precisely those whose information environment official sources dominate.

Aggregating, the mass of runners at fundamental θ is $\sum_k \mu_k \Phi(\sqrt{\beta}(x_k^* - \theta))$ for composition weights μ_k, and the crisis threshold solves the fixed point

$$\theta^* = \sum_k \mu_k \Phi\left(g_k(\theta^*) + \frac{\alpha}{\sqrt{\beta}}\delta_k\right), \qquad g_k(\theta^*) \equiv \frac{\alpha(\theta^* - y) - \Phi^{-1}(\bar{p}_k)\sqrt{\alpha+\beta}}{\sqrt{\beta}}. \tag{4}$$

Proposition 3 (Uniqueness is ambiguity-proof)**.** *If $\alpha/\sqrt{\beta} < \sqrt{2\pi}$, equation (4) has a unique solution, and the threshold equilibrium is the unique equilibrium in monotone strategies, for every composition (μ_k) and every profile of ambiguity radii (δ_k). Ambiguity shifts the map while leaving its slope controlled by the same precision ratio.*

The condition is the familiar one (Hellwig, 2002; Morris and Shin, 2003). Its economic content is independence from δ. Distrust operates like a family of parallel signal translations, so heterogeneous radii preserve the slope condition. The reference precision ratio is 1.35 against the bound 2.51. The multiplier $1/(1-\Phi')$ varies with the location of marginal creditors. Across the unique-equilibrium design distribution its median is 1.56, its interquartile range is 1.30 to 2.07, and its 90th percentile is 2.66.

On a twelve-cell sensitivity grid, private noise ranges from 0.10 to 0.30 and public noise from 0.25 to 0.50. The ratio $\alpha/\sqrt{\beta}$ runs from 0.40 to 4.80, with 3 cells outside the sufficient uniqueness region. At baseline public noise, the frontier for private noise is 0.279. The effective-pessimism stress increment exceeds the calm increment in every unique cell. Across the broader state design, the multiplier ranges from 1.17 at the 10th percentile to 2.66 at the 90th percentile.

Remark 1 (Polarization and source identification)**.** *By Lemma 1, one announcement y enters type k's posterior as $y - \delta_k$. Holding private signals fixed, posterior means differ by $\alpha/(\alpha+\beta)$ times the cross-sectional dispersion of δ_k. Common signal noise leaves this cross-sectional wedge at zero. Heterogeneous Bayesian bias beliefs produce the same wedge, as Proposition 1 requires. Survey disagreement around releases therefore disciplines the distribution of effective pessimism. Preference evidence identifies its source.*

What the static game leaves open is who the types are and how many of each hold the debt. That composition is the paper's state variable, and the next section makes it the comparative-static of interest.

3 The Investor Base

For exposition, group creditors into low-run-propensity and high-run-propensity types, indexed by S and F, with weights $1-\mu$ and μ. Their payoff hurdles $\bar{p}_S$ and $\bar{p}_F$ may differ. The low type has effective pessimism δ_S, while the high type has $\delta_F \geq \delta_S$. The quantitative benchmark sets common payoffs and $\delta_S = 0$ to isolate the information margin. The empirical implementation estimates each sector's payoff hurdle from its funding structure, regulatory treatment, recovery exposure, and observed exit behavior. These estimated hurdles determine run propensity within holder sectors.

Proposition 4 (Composition is priced, with amplification and interaction)**.** *At the unique equilibrium of* (4)*, define $A_k = g_k + (\alpha/\sqrt{\beta})\delta_k$. The composition derivative is*

$$\frac{d\theta^*}{d\mu} = \frac{\Phi(A_F) - \Phi(A_S)}{1 - \Phi'}.$$

Here $\phi_k = \phi(A_k)$ and $\Phi' = (\alpha/\sqrt{\beta})\sum_k \mu_k \phi_k$. It is positive if and only if the high type is more likely to run at the threshold. Holding payoff hurdles fixed, a rise in δ_F raises the crisis threshold. The size of either effect varies with the normal densities at the two run margins and with the coordination multiplier.

In the common-payoff reference calculation, effective pessimism raises the spread by 18 basis points in calm news and 98 basis points in stress. Proposition 1 assigns these prices to the common equilibrium effect of max-min ambiguity and Bayesian pessimistic bias.

The numerator is the direct difference in run propensities. The equilibrium consequence is larger by $1/(1-\Phi')$. Forecast microdata discipline the direct response and the signal precisions. The fixed-point derivative then determines the implied amplification within the model.

4 Official Information: Precision and Credibility

An official communication policy in this model is a pair (α, δ): how sharp the signal is, and how far it can be doubted. The pair moves the crisis threshold through different doors, and the distinction carries all of the section's content.

Proposition 5 (Precision stabilizes conditionally, credibility unconditionally)**.** *(i) The effect of official precision on the crisis threshold satisfies*

$$\operatorname{sign}\left(\frac{d\theta^*}{d\alpha}\right) = \operatorname{sign}\left(y^\dagger - y\right), \qquad y^\dagger = \theta^* + \frac{\sum_k \mu_k \phi_k \left[\delta_k - \Phi^{-1}(\bar{p}_k)/(2\sqrt{\alpha+\beta})\right]}{\sum_k \mu_k \phi_k},$$

so sharper official information lowers the crisis threshold if and only if $y > y^\dagger$. The hurdle exceeds θ^* when every type has $\bar{p}_k < 1/2$. A rise in a type's distrust raises this hurdle in proportion to its mass and its density at the run margin. (ii) Credibility has an unambiguous threshold effect, $d\theta^*/d\delta_k = \mu_k \phi_k (\alpha/\sqrt{\beta})/(1-\Phi') > 0$, for every news level and every composition whenever $\alpha > 0$.

Precision raises the weight placed on the announcement and on the common component of beliefs. It helps after sufficiently favorable news and hurts after sufficiently unfavorable news. Distrust raises the amount of good news required for precision to help. A reduction in δ_k lowers the threshold at every announcement. At the benchmark, halving the high type's radius lowers the stress spread by 50 basis points.

4.1 Ex ante welfare of official precision

Let $L(\theta) > 0$ denote the social loss from a rollover crisis and let $C(\alpha)$ denote the resource cost of producing a signal with precision α. For a proper prior F and the signal density generated by that prior, ex ante welfare is

$$W(\alpha) = -\int L(\theta)\mathbf{1}\{\theta \leq \theta^*(y, \alpha)\} f_\alpha(y \mid \theta)\, dy\, dF(\theta) - C(\alpha). \tag{5}$$

Equation (5) supplies the policy accounting boundary. A precision program raises welfare exactly when its integrated reduction in crisis losses exceeds its resource cost. A credibility reform lowers each δ_k and contracts the crisis region pointwise in the announcement. Its net benefit subtracts the reform cost. The conditional threshold result enters the ex ante policy calculation through this accounting (Angeletos and Pavan, 2007).

The next result separates distrust from a second-moment measure.

Proposition 6 (Distrust moves direction, noise sets the anchor)**.** *(i) $d\theta^*/d\delta_k > 0$ whenever type k has positive mass and $\alpha > 0$. (ii) As official precision vanishes, the threshold converges to*

$$\theta^*_\infty = \sum_k \mu_k(1 - \bar{p}_k).$$

The complete removal of public precision changes the threshold by $\theta^*_\infty - \theta^*$. It is destabilizing when the current threshold is below the anchor and stabilizing when the threshold is above it. (iii) In the latter states, the effect of removing public information and the effect of greater distrust have opposite signs. A variance measure is therefore not a global proxy for distrust.

Figure 3 reports a finite scenario in which the official signal standard deviation rises by 0.08. The distrust shift raises the threshold throughout the plotted news range. The noise experiment changes sign because it moves the threshold toward the no-information anchor. This plotted comparison is a scenario result. The proposition signs the distrust derivative and the limit as public precision vanishes.

5 Dynamics and Policy

The dynamic block links successive rollover games through composition, distrust, and fundamentals news. The calculations below change one slow state at a time.

Quantitative tightening as a composition treatment. A central bank holds maturing debt with zero run margin: it is the limiting stable holder. Balance-sheet runoff hands each maturing bond back to the private margin, mechanically raising μ_t. The experiment in Section 6 drifts μ from 0.35 to 0.65 over eight years, the order of magnitude of post-2022 runoff programs relative to maturing stocks, and prices the drift twice: against calm news throughout, and against a moderate deterioration arriving in year seven. The composition premium is 10 basis points in the first case and 44 in the second. QT's rollover cost is a state-contingent exposure, essentially all of which is collected in bad states. A debt manager who evaluates runoff at calm-market spreads is marking an option position at its premium rather than its payoff.

The doom loop. Higher spreads worsen future fundamentals through debt service, which feeds back into y_{t+1}. Appendix C shows the feedback adds a genuinely dynamic instability: past a spread level, the stable fixed point disappears and the sovereign spirals, the Calvo (1988) mechanism in modern dress, quantified for the Bayesian case by Bocola and Dovis (2019). The headline experiments set this feedback to zero to isolate the composition and credibility mechanisms. With a positive feedback, the same wedges shrink the distance between the calm equilibrium and the spiral's basin boundary.

Collective action clauses as coordination policy. CACs and their statutory relatives raise expected recovery $\mathcal{R}$ in restructuring. Through (2) they raise the hurdle $\bar{p}$, making creditors tolerant of more crisis risk before running, and through Proposition 6(ii) they also lower the uninformative-signal anchor $1-\bar{p}$. Both margins are quantified in Section 6: recovery of 0.70 rather than 0.60 cuts the stress spread from 575 to 128 basis points, and shrinks the ambiguity premium itself, from 0.036 to 0.033 in threshold units, because bounding the loss bounds what the worst case can threaten. That last effect is the model's precise version of the proposal's conjecture that contractual design matters more in ambiguous environments: design that compresses worst-case payoffs is a substitute for trust.

Backstops as ambiguity floors. An official facility that commits to lend against fundamentals above some floor truncates the support of the worst case: a creditor may doubt the budget numbers, while the facility itself anchors the lower tail. In multiple-priors terms the backstop cuts δ directly, which by Proposition 5(ii) stabilizes unconditionally, and its value is largest exactly when δ is high, a reading of catalytic finance (Morris and Shin, 2006) in which the catalyst works on the prior set rather than the signal. The corollary for program design is uncomfortable but useful: a backstop whose own activation is discretionary re-introduces the ambiguity it was meant to remove, and prices accordingly.

Table 1: Quantitative scenario.

Parameter		Value	Category	Basis
Rollover yield	R	1.04	scenario	promised gross return
Recovery	$\mathcal{R}$	0.60	external	haircut evidence (Cruces and Trebesch, 2013)
Exit cost	κ	0.08	scenario	exit payoff
Hurdle	$\bar{p}$	27.3%	internal	equation (2)
Private noise	σ_x	0.15	scenario	forecast-microdata target
Official noise	σ_y	0.33	scenario	real-time-error target
Precision ratio	$\alpha/\sqrt{\beta}$	1.35	internal	uniqueness bound 2.51
Effective pessimism	δ	0.08	scenario	revision history and belief response
High-run type weight	μ	0.40	scenario	holdings and estimated payoff hurdle
News, calm / stress	y	1.10 / 0.95	scenario	capacity units

Table 2: Effective-pessimism-equivalent spread increments.

	Calm ($y = 1.10$)		Stress ($y = 0.95$)	
	Bayesian	Distrust	Bayesian	Distrust
Crisis threshold θ^*	0.377	0.412	0.549	0.584
Crisis probability (%)	1.5	2.0	11.5	13.6
Spread (bp)	61	79	477	575
Effective-pessimism increment (bp)	18		98	

Notes: $\mu = 0.40$ and $\delta = 0.08$. "Bayesian" sets $\delta = 0$. Proposition 1 gives the same differences under max-min ambiguity and Bayesian pessimistic-bias beliefs.

6 Quantitative Magnitudes

6.1 Calibration

Table 1 defines the reference calculation. Recovery of 0.60 follows the average haircut evidence in Cruces and Trebesch (2013). The promised return and exit cost imply the rollover hurdle. Forecast microdata, release responses, and holder behavior supply the empirical targets for the information and investor-base block. The reference point satisfies the uniqueness condition. A broader design distribution spans 1,836 unique-equilibrium states and reports the full multiplier distribution.

6.2 Results

Table 2 reports the reference arithmetic. The same effective pessimism raises spreads by 18 basis points in calm news and 98 basis points in stress because more creditors lie near the run margin after adverse news. The coordination multiplier varies across the state space. Its 10th percentile is 1.17, its median is 1.56, its 90th percentile is 2.66, and the maximum unique-equilibrium value is 22.66. Figure 1 traces thresholds and stress spreads across investor composition. The remaining figures report the precision boundary, the distinction between distrust and noise, and the composition path.

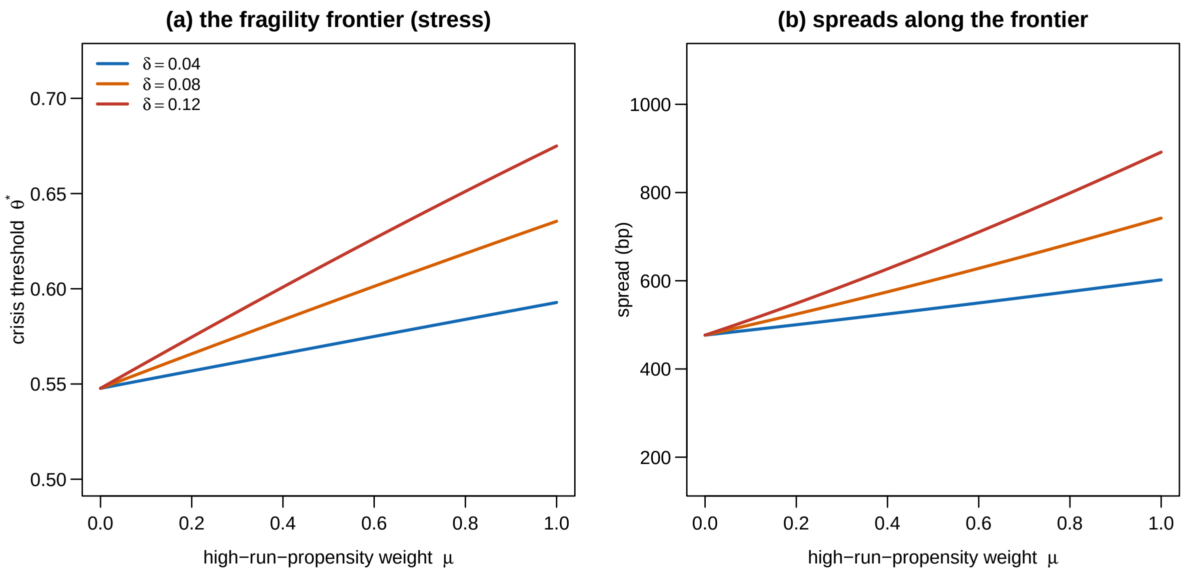

Figure 1: The composition frontier under stress news: crisis threshold (a) and spread (b) by the high-run-propensity type's weight, for effective-pessimism radii $\delta \in \{0.04, 0.08, 0.12\}$.

7 Empirical Designs

The empirical work first measures official errors, creditor beliefs, and type-specific rollover incentives. Market outcomes then test the coordination mechanism in Proposition 4.

Revision histories and subjective pessimism. Let y_{cvt} be a real-time official forecast for country c, vintage v, and target date t. Put each forecast in liquid-capacity units using a predetermined mapping from debt service, fiscal balances, and reserves. The revision $e_{cvt} = y_{cvt} - y_{cFt}$ compares the release with the final vintage. Its upper quantiles describe realized overstatement. The following measurement equation maps that history and direct beliefs into δ_k:

$$\delta_{kct} = h_k(Q_c(e), Z_{kct}), \tag{6}$$

where Z_{kct} contains direct survey beliefs and each creditor type's forecast response to official releases. The IMF historical WEO archive supplies public vintages (International Monetary Fund, 2026). ECB SPF and Consensus forecast panels reveal how professional forecasters discount an official release after past revisions. The estimated h_k maps objective error history into subjective effective pessimism.

Signal precision. Forecast dispersion combines private information with persistent models, sticky updating, loss functions, and rounding. A hierarchical measurement model separates these

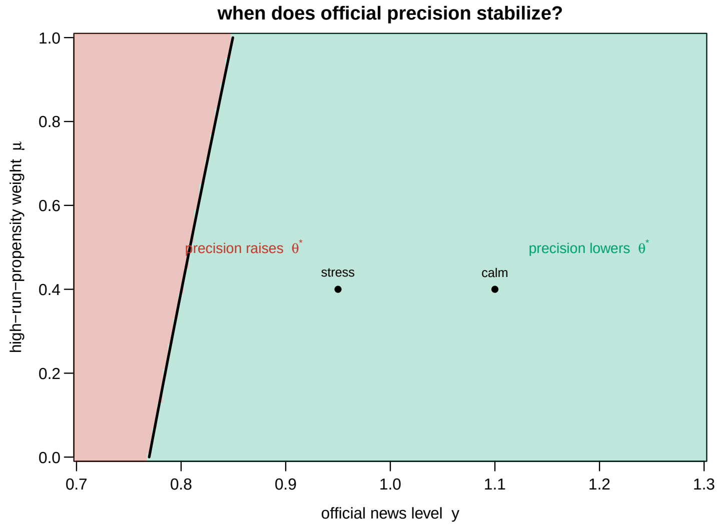

Figure 2: The sign of $d\theta^*/d\alpha$: official precision stabilizes in the green region and destabilizes in the red, separated by the boundary $y^\dagger(\mu)$ of Proposition 5. Dots mark the calm and stress calibrations at $\mu = 0.40$.

components:

$$\begin{aligned} f_{jcrh} &= x_{jcrh} + a_j + \ell_{jh} + m_{jc} + \zeta_{jc,r-1} + u_{jcrh}, \\ x_{jcrh} &= \theta_{crh} + \varepsilon_{jcrh}, \qquad \varepsilon_{jcrh} \sim N(0, \sigma_x^2). \end{aligned} \tag{7}$$

Forecaster effects a_j, horizon effects ℓ_{jh}, persistent model components m_{jc}, and lagged-information states $\zeta_{jc,r-1}$ absorb systematic disagreement. An interval-censored likelihood treats rounded reports. Repeated forecasts identify persistence separately from the transitory private signal. Real-time official forecast errors identify σ_y^2. ECB SPF microdata provide public individual forecasts (European Central Bank, 2026b), while licensed Consensus forecasts extend the country panel. Posterior draws for σ_x^2 and σ_y^2 enter the equilibrium solver jointly.

Investor composition and payoff hurdles. Holder accounts reveal creditor sectors, while release behavior estimates run propensity. ECB Securities Holdings Statistics by Sector provide current euro-area weights μ_{kct} (European Central Bank, 2026a). World Bank QPSD and QEDS provide current debt stocks and creditor splits (World Bank, 2026b; World Bank, 2026a). Sector flows around adverse releases, auction participation, funding maturity, and regulatory liquidity treatment identify $\bar{p}_k$. The empirical aggregate run index is

$$M_{ct} = \sum_k \mu_{kct} \Phi(\widehat{A}_{kct}^{;0}), \tag{8}$$

where $\widehat{A}_{kct}^{;0}$ sets effective pessimism to zero and retains the estimated payoff hurdle. This construction keeps the composition regressor separate from D_{ct} and preserves continuous heterogeneity. The two-type quantitative figure projects this estimated distribution onto low and high run

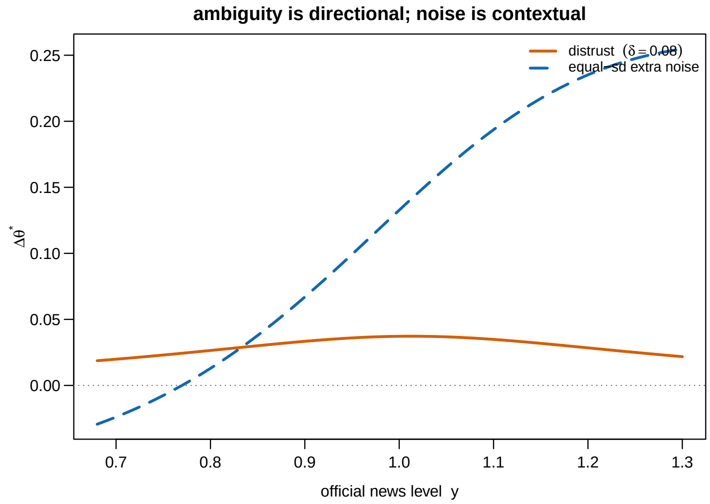

Figure 3: Finite scenario comparison. The solid line gives the threshold effect of $\delta = 0.08$. The dashed line gives the effect of raising the official signal standard deviation by 0.08.

propensities. The historical panel of Arslanalp and Tsuda (2014) covers its original sample period.

Market test. With $(\delta, \alpha, \beta, M)$ measured before market outcomes, the core quarterly specification is

$$\Delta s_{i,t} = \alpha_i + \tau_t + \beta_1 D_{i,t} + \beta_2 M_{i,t} + \beta_3 \text{Stress}_{i,t} + \beta_4 D_{i,t} M_{i,t} + \beta_5 D_{i,t} M_{i,t} \text{Stress}_{i,t} + \Gamma' X_{i,t} + \varepsilon_{i,t}, \quad (9)$$

where D is the effective-pessimism estimate from (6) and stress uses information dated before the spread change. Proposition 4 predicts $\beta_4 > 0$ when composition loads on higher run propensities. Density at the run margin predicts $\beta_5 > 0$. Forecast-variance posterior draws enter separately, preserving the distinction between pessimism and noise.

Primary auctions provide the quantity counterpart. Bid-to-cover, allotment tails, and cancellations are projected on the same predetermined interaction with issue fixed effects and narrow announcement windows. These outcomes reveal rollover pressure more directly than secondary-market spreads. Public release dates also permit tests of the sign boundary in Proposition 5.

Institutional reforms provide descriptive validation. Dissemination standards, fiscal councils, Article IV publications, and program reviews follow selected policy calendars. Their event windows show whether forecast responses and market sensitivity move in the direction of lower δ_k. Equation (9), estimated from predetermined information and holder states, remains the core test.

Table 3 states model restrictions and the data moment that bears on each. Forecast polarization and release responses estimate effective pessimism. Choice evidence under known bias distributions determines whether that pessimism reflects max-min preferences or Bayesian beliefs.

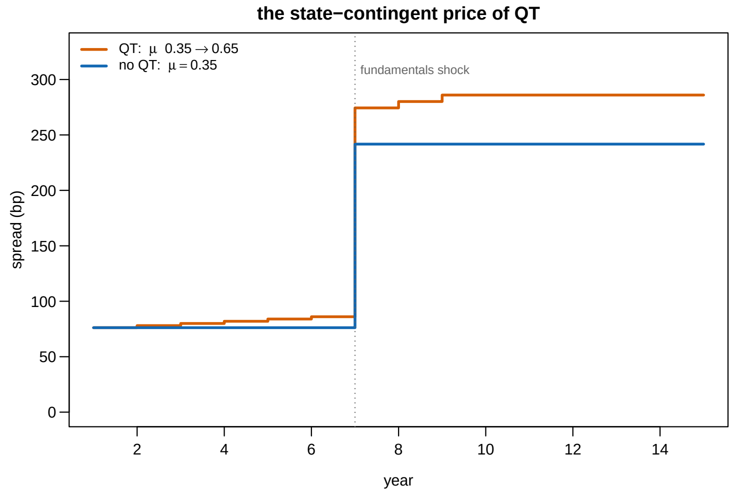

Figure 4: The state-contingent price of quantitative tightening: spread paths as the high-run-propensity weight moves from 0.35 to 0.65, against a fixed-weight counterfactual. A moderate fundamentals shock arrives in year seven.

8 Conclusion

The investor base determines how official distrust enters a rollover crisis. Effective pessimism lowers type k's public signal by δ_k and raises its run cutoff by $(\alpha/\beta)\delta_k$. Type-specific payoff hurdles locate marginal creditors. Their portfolio weights aggregate those margins into one crisis threshold, while $1/(1-\Phi')$ measures coordination amplification.

The same equilibrium follows from max-min ambiguity and Bayesian pessimistic bias. This equivalence organizes the evidence. Revision histories measure objective reporting errors. Creditor beliefs map those errors into δ_k. Forecast panels identify information precision through the hierarchical measurement model. Holder accounts and observed exits identify the distribution of payoff hurdles. The triple interaction in equation (9) then tests the market mechanism.

Official precision changes sign with news, while credibility lowers the crisis threshold at every announcement. The reference spread increment rises from 18 basis points in calm news to 98 basis points in stress. Across the unique-equilibrium state space, the multiplier has median 1.56 and 90th percentile 2.66. These results show why a debt manager must price the creditor base together with the information environment.

Table 3: Predictions and implementation.

Prediction	Model magnitude	Null	Estimation
Effective pessimism interacts with the run-propensity index	$\beta_4 > 0$ in eq. (9)	additive effects	spread panel
The cutoff response scales with official-information dependence	loading α/β	uniform	forecast microdata
Composition shifts raise sensitivity to adverse news	10 calm, 44 stress	state invariant	holder-sector panel
Precision changes sign at $y^{\dagger}$, credibility does not	Figure 2	uniform sign	release windows
Complete information loss converges to the weighted payoff anchor	Proposition 6	current threshold	vintage comparisons
Effective pessimism widens forecast disagreement	Remark 1	common response	individual forecasts

Table 4: Data and implementation.

Dataset	Coverage	Freq.	Access	Status	Use
ECB SHSS by sector (European Central Bank, 2026a)	euro-area securities	quarterly	public aggregates	observed	current holder shares
World Bank QPSD and QEDS (World Bank, 2026b; World Bank, 2026a)	reporting economies	quarterly	public	observed	current debt stocks
Historical investor base (Arslanalp and Tsuda, 2014)	AE and EM sample	quarterly	public	observed	historical panel
Real-time official forecasts (International Monetary Fund, 2026)	WEO reporters	2/y	public	observed	revision histories
Data-standard subscriptions	IMF dissemination standards	event	public	observed	event panel
ECB SPF microdata (European Central Bank, 2026b)	euro area	quarterly	public	individual	precision moments
Consensus forecasts	country sample	monthly	licensed	individual	precision moments
Bond and CDS spreads	liquid sovereigns	daily	licensed	observed	licensed panel
Auction microdata	debt-office releases	per auction	uneven	observed where released	debt-office sample
Institutional and release dates	official archives	event	public	observed	timing tests

A Proofs

A.1 Proof of Lemma 1

Fix a profile of cutoff strategies. Its run mass is continuous and strictly decreasing in θ, while liquid capacity is strictly increasing. There is one crossing θ^* and crisis occurs exactly when $\theta \leq \theta^*$. Under bias model b, let $P_b = \mathbb{P}_b(\theta \leq \theta^* \mid x_i, y)$. The posterior mean is $[\alpha(y - b) + \beta x_i]/(\alpha + \beta)$, so P_b is increasing in b. The expected payoffs are

$$U_k^R(b) = R_k - P_b(R_k - \mathcal{R}_k), \qquad U_k^E(b) = Q_k - P_b(Q_k - q_k).$$

Assumption 1 makes both functions weakly decreasing in P_b. Their minima are therefore attained at $b = \delta_k$. Comparing the two minima gives

$$U_k^R(\delta_k) - U_k^E(\delta_k) = R_k - Q_k - P_{\delta_k}\left[(R_k - \mathcal{R}_k) - (Q_k - q_k)\right].$$

The creditor rolls precisely when $P_{\delta_k} \leq \bar{p}_k$. This is the Bayesian decision computed from effective signal $y - \delta_k$. ■

A.2 Proof of Proposition 1

Lemma 1 maps every type's max-min problem into a Bayesian problem with the same payoff hurdle and the effective public signal $y - \delta_k$. Fix any cutoff profile. The mapping gives the same action for every private signal, hence the same aggregate run function. Its crossing with liquid capacity is the same. Applying the argument to the equilibrium cutoff profile proves equality of equilibrium choices and crisis probabilities. Any price that is a fixed function of the equilibrium crisis probability and state-contingent payoffs is also the same. ■

A.3 Proof of Proposition 2

With posterior $\theta \mid x \sim N(m_k(x), (\alpha + \beta)^{-1})$, indifference requires

$$m_k(x_k^*) = \theta^* - \frac{\Phi^{-1}(\bar{p}_k)}{\sqrt{\alpha + \beta}}.$$

Since $m_k(x) = [\alpha(y - \delta_k) + \beta x]/(\alpha + \beta)$, solving for the cutoff gives $x_k^* = x_{B,k}^* + (\alpha/\beta)\delta_k$, where $x_{B,k}^*$ is the cutoff at $\delta_k = 0$ with the same payoff hurdle. Multiplication by $\sqrt{\beta}$ gives the shift $(\alpha/\sqrt{\beta})\delta_k$ in the run index. ■

A.4 Proof of Proposition 3

Substituting the cutoffs into the crossing condition gives (4). Its right side has derivative

$$\frac{\alpha}{\sqrt{\beta}} \sum_k \mu_k \phi\left(g_k + \frac{\alpha}{\sqrt{\beta}}\delta_k\right) \leq \frac{\alpha}{\sqrt{2\pi\beta}} < 1.$$

The bound is uniform in payoffs, composition, and distrust. The map sends $[0, 1]$ into $(0, 1)$ and is a contraction, so it has one fixed point. Each type's posterior crisis probability is strictly decreasing in its private signal, which gives one cutoff for that threshold. The resulting profile is the unique monotone equilibrium. ■

A.5 Proof of Proposition 4

Let $s = \alpha/\sqrt{\beta}$, $A_F(\theta) = \Phi(g_F(\theta) + s\delta_F)$, and $A_S(\theta) = \Phi(g_S(\theta) + s\delta_S)$. With two types, the fixed point is

$$H(\theta, \mu) = (1 - \mu)A_S(\theta) + \mu A_F(\theta) - \theta = 0.$$

Write $\Phi' = s[(1 - \mu)\phi_S + \mu\phi_F]$. The uniqueness condition gives $H_\theta = \Phi' - 1 < 0$. Implicit differentiation gives

$$\frac{d\theta^*}{d\mu} = -\frac{H_\mu}{H_\theta} = \frac{A_F(\theta^*) - A_S(\theta^*)}{1 - \Phi'}.$$

Its sign is the sign of the difference in the two run probabilities. Holding $\bar{p}_F$ fixed, differentiation with respect to δ_F gives

$$\frac{d\theta^*}{d\delta_F} = \frac{\mu s\phi_F}{1 - \Phi'} > 0.$$

The numerator densities locate the marginal creditors. The denominator is the coordination multiplier. Both vary with the state. ■

A.6 Proof of Proposition 5

Differentiate (4) in α. Holding θ^* fixed gives

$$\frac{\partial}{\partial\alpha}\left(g_k + \frac{\alpha}{\sqrt{\beta}}\delta_k\right) = \frac{(\theta^* - y) - \Phi^{-1}(\bar{p}_k)/(2\sqrt{\alpha + \beta}) + \delta_k}{\sqrt{\beta}}.$$

Implicit differentiation therefore gives

$$\frac{d\theta^*}{d\alpha} = \frac{\sum_k \mu_k\phi_k\left[(\theta^* - y) - \Phi^{-1}(\bar{p}_k)/(2\sqrt{\alpha + \beta}) + \delta_k\right]}{\sqrt{\beta}(1 - \Phi')},$$

whose sign is that of $y^\dagger - y$. If every $\bar{p}_k < 1/2$, every quantile term is positive. The distrust terms are nonnegative. Hence $y^\dagger > \theta^*$. Direct differentiation in δ_k gives part (ii). ■

A.7 Proof of Proposition 6

Part (i) follows from direct differentiation of (4). As $\alpha \to 0$, $g_k \to -\Phi^{-1}(\bar{p}_k)$ and the distrust shifts vanish. The right side converges uniformly on $[0, 1]$ to

$$\sum_k \mu_k\Phi[-\Phi^{-1}(\bar{p}_k)] = \sum_k \mu_k(1 - \bar{p}_k).$$

Uniform convergence and uniqueness imply convergence of the fixed point to this constant. Subtracting the current threshold gives the finite effect of removing public precision. Part (iii) compares that signed difference with the strictly positive derivative in part (i). ■

B Derivations: Pricing and Pass-Through

Spreads. With risk-neutral pricing at a zero reference rate, a bond promising R with crisis probability P and recovery $\mathcal{R}$ carries spread $s = P(1-\mathcal{R})/[1-P(1-\mathcal{R})]$ per period, the mapping used throughout. For small P it is $P\times$ loss given default. Endogenizing R through an issuance condition adds a second fixed point that raises all fragility statics (a higher threshold raises the promised yield, which through $\bar{p}$ raises the threshold again). The reported magnitudes are conservative in omitting it.

News pass-through and the multiplier. Differentiating (4) in y gives $d\theta^*/dy = -\Phi'/(1-\Phi')$ and hence

$$\frac{dP}{dy} = \phi\big((\theta^* - y)\sqrt{\alpha}\big)\sqrt{\alpha}\left(\frac{d\theta^*}{dy} - 1\right) = -\frac{\phi\big((\theta^* - y)\sqrt{\alpha}\big)\sqrt{\alpha}}{1-\Phi'}\ :$$

the coordination multiplier scales the pass-through of public news to crisis risk. Forecast moments discipline the direct density and fixed-point slope before the empirical implementation maps this expression into market data.

C Dynamics and the Doom Loop

Composition evolves by accounting: with runoff fraction q_t of the official portfolio's maturing stock, $\mu_{t+1} = \mu_t + q_t(1-\mu_t)\omega_t$, where ω_t is the official share of maturing debt. The QT experiment in the text uses a linear drift matching an eight-year normalization. The debt-service feedback closes fundamentals: $y_{t+1} = \bar{y}_{t+1} - \varsigma\,(s_t - s^{ref})$, service costs subtracting from next period's capacity news. Combining with the pass-through result above, the loop is locally explosive when

$$\varsigma \times \frac{\phi\big((\theta^* - y)\sqrt{\alpha}\big)\sqrt{\alpha}\,(1-\mathcal{R})}{1-\Phi'}\ >\ 1,$$

a condition that tightens mechanically as stress raises the density term and as composition and distrust raise Φ': the same statics that price spreads shrink the distance to the spiral. This is the Calvo (1988) instability with the coordination multiplier made explicit. Bocola and Dovis (2019) quantify the Bayesian analogue. The text's experiments hold $\varsigma = 0$ to isolate the state-contingent spread gap that any positive ς would amplify.

D Parameter Discipline and Sensitivity

Table 5 records where each parameter enters, its units, its source, the range examined, and what moves. At $\delta = 0$ the model is the standard Bayesian global game and the distinctive predictions

P1, P4's distrust term, P5, and P6 return to zero. As $\alpha/\sqrt{\beta}$ approaches 2.51, the multiplier grows and the distance to selection becomes the relevant statistic, with the margin discussed in Section 2.

Table 5: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source or moment	Range examined	Effect of the range
R, κ	eq. (2)	face value	gross yield, fire-sale discount	joint shift	shift $\bar{p}$ jointly
$\mathcal{R} = 0.60$	eq. (2)	recovery rate	Cruces and Trebesch (2013)	0.60 to 0.70 (CAC exp.)	stress spread 575 to 128 bp
$\bar{p} = 27.3\%$	eq. (4)	probability	implied	follows $\mathcal{R}$	anchor $1 - \bar{p}$ moves
σ_x, σ_y	eqs. (4), (7)	capacity units	hierarchical forecast model	σ_x 0.10 to 0.30, σ_y 0.25 to 0.50	multiplier p10 1.17, median 1.56, p90 2.66
$\alpha/\sqrt{\beta} = 1.35$	uniqueness	unitless	implied	up to bound 2.51	multiplier grows, then multiplicity
$\delta = 0.08$	eqs. (3), (6)	capacity units	error history and belief response	0.04 to 0.12	spread increment 18 to 98 bp
μ_k, $\bar{p}_k$	eqs. (4), (8)	debt share, probability	holder accounts and observed exits	full type distribution	QT gap 10 to 44 bp
y calm/stress	eq. (4)	fundamentals units	internal, calm spread 79 bp	full axis in figures	state dependence of every result

Notes: recovery follows external haircut evidence. The reference point sits inside the ranges shown. Section 7 defines the measurement equations for the information and holder blocks.

E Data Appendix

Investor base. ECB Securities Holdings Statistics by Sector supply current euro-area holder weights (European Central Bank, 2026a). World Bank QPSD provides current public debt stocks (World Bank, 2026b). QEDS provides external debt statistics for reporting economies (World Bank, 2026a). The data of Arslanalp and Tsuda (2014) supply their historical panel. Debt-office maturity schedules align holdings with the rollover date. Sector flows and auction participation estimate payoff hurdles, producing the continuous run index in (8).

Effective pessimism. The IMF historical WEO archive supplies the public vintage panel (International Monetary Fund, 2026). Revision quantiles measure realized reporting errors. Equation (6) combines them with forecast responses and direct belief evidence to estimate δ_k. Data-standard subscriptions (Cady, 2004) and documented statistical-integrity events validate the direction of the estimated mapping.

Information precision. ECB SPF individual forecasts provide public microdata for euro-area expectations (European Central Bank, 2026b). Licensed Consensus forecasts extend the country panel. Repeated observations identify persistent models and sticky updating in (7). Interval censoring absorbs strategic rounding. The remaining transitory component identifies private-signal variance after every outcome enters common capacity units.

Market outcomes. Sovereign bond and CDS spreads at daily frequency for announcement windows and quarterly for the panel. Primary-auction results (bid-to-cover, stop-out tails, cancellations) from debt-management offices, and IMF Article IV and program-review publication dates from the IMF archive.

F Extensions

Heterogeneous private precision. If type k has private precision β_k, its cutoff shift becomes $(\alpha/\beta_k)\delta_k$ and its run index has slope $\alpha/\sqrt{\beta_k}$. A sufficient uniqueness condition is $\sum_k \mu_k \alpha/\sqrt{2\pi\beta_k} < 1$. Forecast microdata can discipline the sector-specific variances. The composition derivative retains the difference in type-specific run probabilities divided by the derivative of the aggregate fixed-point map.

Smooth ambiguity. Replacing max-min with smooth ambiguity aversion turns the worst-case shift δ_k into an endogenous pessimism weight varying with stakes. The as-if structure survives to first order, with δ_k reinterpreted as the local ambiguity premium. The max-min case is the tractable bound.

Learning about bias. Letting creditors update the bias set from realized revisions makes δ_t a state with its own persistence: credibility as reputational capital. The asymmetry between slow accumulation and one-announcement destruction suggests trap dynamics complementary to those studied here.

Dynamic runs. Staggered maturities in the sense of He and Xiong (2012) let today's rollover outcome inform tomorrow's, adding an intertemporal coordination motive to the static one. The composition state μ_t then also indexes the speed at which a run can build. The static threshold is the natural building block for that model.

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